

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of this project is to develop a machine learning model that predicts whether a credit card application will be approved based on an applicant's financial and personal information. The system aims to provide accurate, fast, and reliable approval decisions.
Scope	The project focuses on collecting and preprocessing applicant data, training a machine learning model, and deploying a web-based application that predicts credit card approval in real time. The solution can assist financial institutions in improving efficiency and reducing approval time.
Problem Statement	
Description	Traditional credit card approval systems often require manual verification, making the process time-consuming and prone to human errors. Financial institutions need an automated system that can evaluate applications quickly and accurately.

Impact	Implementing an automated credit card approval prediction system will: • Reduce manual effort. • Improve prediction accuracy. • Speed up application processing. • Support consistent and data-driven decision-making. • Enhance customer satisfaction.
Proposed Solution	
Approach	The proposed solution uses supervised machine learning to analyze applicant details and predict credit card approval. The data is preprocessed, trained using a Random Forest Classifier, and deployed through a Flask web application for real-time prediction.
Key Features	☐ Machine learning-based credit card approval prediction. ☐ Real-time prediction through a web interface. ☐ Data preprocessing and feature scaling. ☐ High prediction accuracy using Random Forest. ☐ User-friendly interface for entering applicant details.



	☐ Fast and reliable approval decision support.
--	--

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor
Memory	RAM specifications	8 GB RAM

Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Python(Scikit-learn, Pandas, NumPy, Joblib)
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Credit Card Approval Dataset CSV Kaggle / UCI Machine Learning Repository